# Population and Economic Growth
# When Malthus Meets Romer

Anson Zhou

The University of Hong Kong

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# One of the Oldest Debates in Economics

- Does a larger population raise or lower living standards?
  - → Malthus (1798): population strains resources ⇒ wages to subsistence
    - ∗ David Ricardo, Paul Ehrlich, William Catton, . . .
  - → Romer (1990): more people ⇒ more ideas ⇒ innovation dominates
    - ∗ Richard Jones, Julian Simon, Chad Jones, Elon Musk, . . .
- Never more consequential: 2bn to 8bn+ in a century, now fertility below replacement across the developed world
  - → "The end of economic growth" – Jones (2022)
- This paper: study the implications of the disagreement itself

# This Paper

- Overlapping generations model with
  - → Fertility choice under altruism
  - → Semi-endogenous growth dynamics
  - → Heterogeneous beliefs about how population matters for productivity
- Mechanism: beliefs affect household fertility choice + aggregate population dynamics shapes evidence
  ⇒ wrong beliefs can persist and delay growth
- For mechanism to work, only need
  1. Population matters for productivity (core of growth models)
  2. Children's living standard matters for fertility choice (core of intergenerational models)

# Findings

- Theoretical results
  - → Fertility ranking between types is state-dependent
  - → Romerians in a Malthusian world generate instability and cycles
  - → Malthusians in a Romerian world generate population and growth traps
- Quantitative assessment
  - → Inherited disagreement delays takeoff by centuries
  - → Information trap if initial Romerian share too small
- Supporting evidence
  - → Belief correlates with fertility in survey data
  - → Complementary evidence from Malthus at Haileybury

# Related Literature & Contribution

- **Fertility & unified growth**: Barro–Becker (1989), Galor (2000, 2011), Hansen–Prescott (2002); demographic transition and sustained growth
- **Scale effects**: Romer (1990), Kremer (1993), Jones (1995, 2022); the sign of the population–technology link is central to long-run projections
- **Heterogeneous beliefs**: Angeletos–La'O (2013), Coibion–Gorodnichenko (2015); climate disagreement (Bakkensen–Barrage 2022)
- **Cultural transmission/entrepreneur & evolutionary selection**: Bisin–Verdier (2001, 2011), Galor–Moav (2002), Galor–Michalopoulos (2012), Mokyr (2016)

**This paper**: disagreement about population scale effect shapes both the demographic composition of beliefs and the evidence available

# Roadmap

- Model setup
- Theoretical results
- Quantitative application
- Supporting evidence

# Model Setup

# Model: Environment

- Discrete-time OLG; continuum of atomistic households
- Time endowment 1; fertility $n$ costs time $\psi(n)$, labor supply $1 - \psi(n)$
- Linear production: $Y_t = A_t L_t$, wage $= A_t$, so

$$c_{j,t} = A_t\big(1 - \psi(n_{j,t})\big)$$

- Population and its adjusting reference:

$$N_{t+1} = \bar{n}_t N_t, \qquad \bar{n}_t = \phi_t n_{R,t} + (1 - \phi_t) n_{M,t}$$

- $\phi_t$ = share of Romerians at time $t$

# TFP Dynamics: The Population Gap

- Productivity growth depends on population gap:

$$\ln A_{t+1} - \ln A_t = \theta^* \, x_t + \eta_{t+1}, \qquad \eta_{t+1} \sim \mathcal{N}(0, \sigma^2)$$
$$x_{t+1} = (1-\mu) x_t + \ln \bar{n}_t, \qquad \mu \in (0, 1]$$

  delivers only level effect of one-time population increase

- Canonical idea production (Jones 1995)

$$\Delta \ln A_{t+1} = \alpha \ln N_t - \omega \ln A_t + \eta_{t+1} \qquad \alpha > 0, \omega \in (0, 1)$$

  → BGP mapping: $g = \alpha \ln \bar{n} \,/\, \omega$ vs. $g = \theta^* \ln \bar{n} \,/\, \mu$, so $\theta^*/\mu \;\leftrightarrow\; \alpha/\omega$

- Disagreement: $\theta^* \in \{\theta_R, \theta_M\}$, $\theta_M < 0 < \theta_R$ but unknown to households

# Preferences: Barro–Becker Dynasties

why these preferences

- Dynastic utility:

$$V_{jt} = \max_{n_{jt}} \; u\big(c_{jt}\big) + \beta\, n_{jt}^{1-\varepsilon}\, \mathbb{E}_j\big[V_{j',t+1}\big], \qquad \rho, \varepsilon \in (0,1)$$

  subject to $c_{jt} = A_t(1 - \psi(n_{jt}))$

- Time cost of childbearing: increasing and convex
- CRRA utility + time costs $\Rightarrow$ TFP level cancels from the fertility FOC
- Beliefs matter only via expected growth

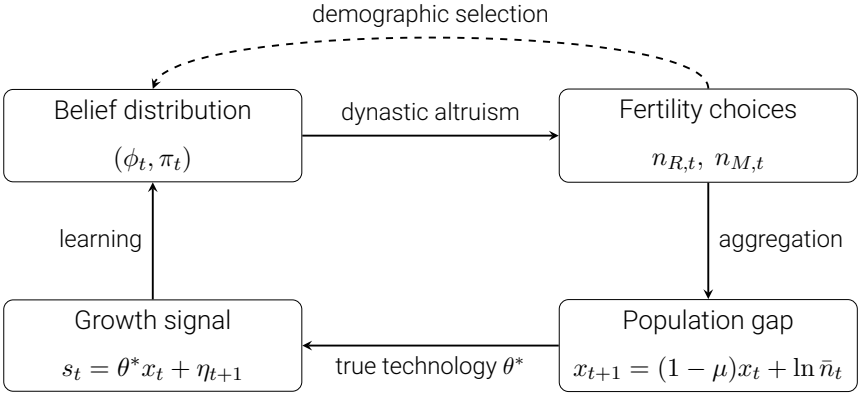
# Beliefs, Learning, and Transmission

- Two types: share $\phi_t$ Romerians ($\theta_R > 0$), $1 - \phi_t$ Malthusians ($\theta_M < 0$)
- Social credence $\pi_t$: shared posterior (the body of knowledge in the society), Bayes-updated from growth signals $s_t = \Delta A_t = \theta^* x_t + \eta_{t+1}$
- Transmission: child inherits parent's worldview with prob $\lambda$; otherwise adopts the Romerian view with prob $\pi_{t+1}$
- Belief distribution evolves via two channels:

$$\phi_{t+1} = \underbrace{\lambda\, \phi_t^{birth}}_{\text{cultural transmission}} + \underbrace{(1-\lambda)\, \pi_{t+1}}_{\text{learning-based recruitment}}$$

- Reproductive selection and social learning can conflict

# The Belief–Fertility–Growth Feedback Loop

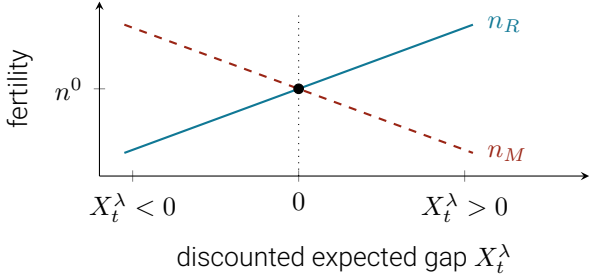



- Near $x = 0$: both feedback channels into beliefs shut down together
- Household choices shape the evidence that future generations use to revise beliefs

# Theoretical Results

# State-Dependent Fertility Ranking

$$n_{R,t} - n_{M,t} \approx \underbrace{\frac{1-\rho}{1+\varepsilon}}_{K} \cdot \Delta\theta \cdot X_t^{\lambda} \quad \text{where} \quad X_t^{\lambda} \equiv \mathbb{E}_t\left[\sum_{k\geq 0}(\delta\lambda)^k x_{t+k}\right]$$




- $X_t^{\lambda} > 0$ (above reference on balance): Romerians out-reproduce
- $X_t^{\lambda} < 0$ (below reference on balance): ranking flips
- At $X_t^{\lambda} = 0$: identical fertility despite disagreement

# Demographic Selection and Learning

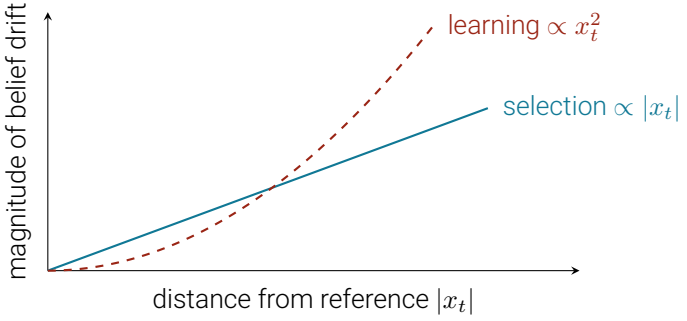
- Expected change in adult Romerian share:

$$\mathbb{E}_t[\phi_{t+1} - \phi_t] = \underbrace{\lambda \frac{\phi_t(1-\phi_t)(n_{R,t} - n_{M,t})}{\bar{n}_t}}_{\text{reproductive selection}} + \underbrace{(1-\lambda)(\pi_t - \phi_t)}_{\text{recruitment to credence}} + \underbrace{(1-\lambda)\mathbb{E}_t[\pi_{t+1} - \pi_t]}_{\text{new evidence}}$$

- Selection favors trajectory-optimists; learning favors the truth

| Expected path | Malthusian world | Romerian world |
|---|---|---|
| Above reference | Selection: Romerians × | Selection: Romerians ✓ |
| | Learning: Malthusians ✓ | Learning: Romerians ✓ |
| Below reference | Selection: Malthusians ✓ | Selection: Malthusians × |
| | Learning: Malthusians ✓ | Learning: Romerians ✓ |

# Two Engines of Belief Change



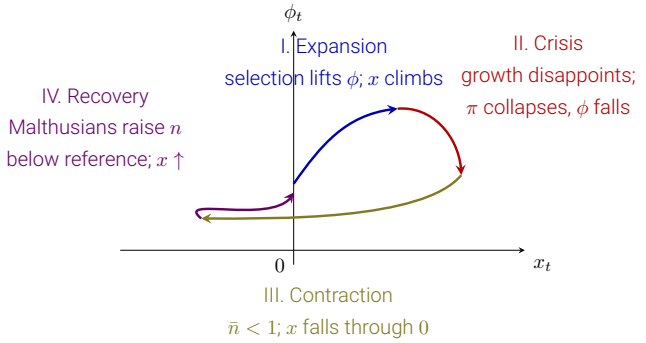

- Selection favors trajectory-optimists — vanishes linearly with $|x_t|$
- Learning favors the truth — vanishes quadratically
- Near reference: selection dominates; far away: learning dominates
- At $x = 0$: both exactly zero — disagreement is self-sustaining

# Information Accumulation

- Cumulative signal information: $\mathcal{C}_T = \sum_{t=0}^{T-1} \mathcal{I}_t \propto \sum_{t=0}^{T-1} x_t^2$
- Theorem (Information and posterior convergence):
  - → On $\{\sum_t \mathcal{I}_t = \infty\}$: social credence $\pi_t \to$ truth a.s.
  - → On $\{\sum_t \mathcal{I}_t < \infty\}$: $\pi_t \to$ interior limit $\pi_\infty \in (0, 1)$
- Sustained fertility above or below replacement ⇒ gap stays away from zero ⇒ infinite info ⇒ truth identified
- Geometric return to reference ⇒ $\sum x_t^2 < \infty$ ⇒ finite info ⇒ persistent disagreement – c.f., robust misspecified models (Ba 2026)
- Whether learning resolves the debate depends on the population path

# Malthusian World: Boom-Bust Cycles

cycle mechanics




- Selection and learning oppose above the reference $\Rightarrow$ tug-of-war
- Busts reinforced by selection; recoveries engineered by pessimists
- Similar in spirit to endogenous business cycles (Beaudry et al. 2020):
  - $\rightarrow$ strategic complementarity (among Romerians)
  - $\rightarrow$ some accumulation variable that when high, tends to depress the individual value of accumulating more (population in this case)

# Romerian World: Dormancy and Escape

trap regimes

- Above reference: selection and learning aligned — both favor Romerians
  $\Rightarrow$ self-cementing escape and growth
- Below or near reference:
  - $\rightarrow$ Exact: $x_0 = 0 \Rightarrow x_t = 0, n_{R,t} = n_{M,t} = 1, \pi_t = \pi_0$ forever; wrong beliefs never corrected
  - $\rightarrow$ Near-dormant: for $|x_0|$ small, with high probability:

$$|x_t| \leq q^t |x_0|, \qquad \sum_t \mathcal{I}_t < \infty, \qquad \pi_t \to \pi_\infty \in (0,1)$$

Population converges to a finite level while leaving the scale-effect disagreement unresolved

# Quantitative Application

# Application: Delayed Takeoff

- Setup: the dawn of the industrial revolution when history supports Malthusian beliefs (Clark 2007)
- Potential regime change (Hansen-Prescott 2002)
- Model: people might not be aware of it
    1. How long could the initial pessimism delay growth take-off?
    2. How large is the welfare consequences of imperfect information?
    3. How large is the role of luck or cultural entrepreneurship (Mokyr 2016)?

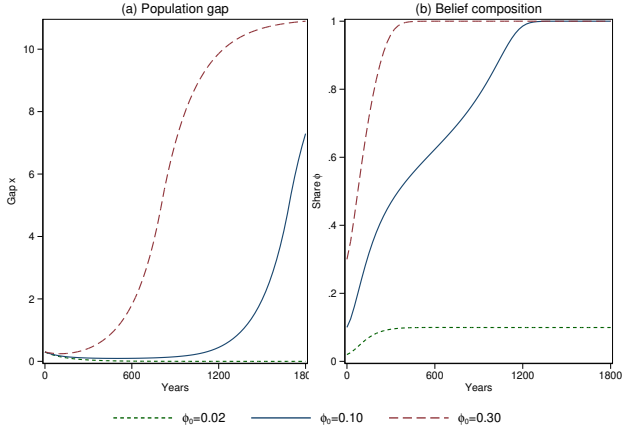
# Calibration

computation details

| Parameter | Value | Target / source |
|---|---|---|
| $\rho, \varepsilon$ | 0.50 | CRRA & dynastic curvature (Jones–Schoonbroodt) |
| $\gamma$ | 0.312 | 15% child-rearing time at replacement |
| $\beta$ | 0.252 | Replacement normalization $\Rightarrow \delta \simeq 0.253$ |
| $\sigma$ | 0.20 | ≈4% annual s.d. of productivity |
| $\mu$ | 0.10 | Gap half-life ≈165 years (Jones 1995, 2022) |
| $\theta_R$ | 0.50 | Multiplier $\theta_R/\mu = 5$ (Kremer 1993); 2%/yr at entry |
| $\theta_M$ | $-0.15$ | Perceived elasticity $-\theta_M/\mu = 1.5$ (Lee 1980, Clark 2007) |
| $\lambda$ | 0.60 | Vertical transmission (Bisin–Verdier 2001, 2011) |
| $x_0$ | 0.30 | Initial displacement |

# Threshold Dynamics

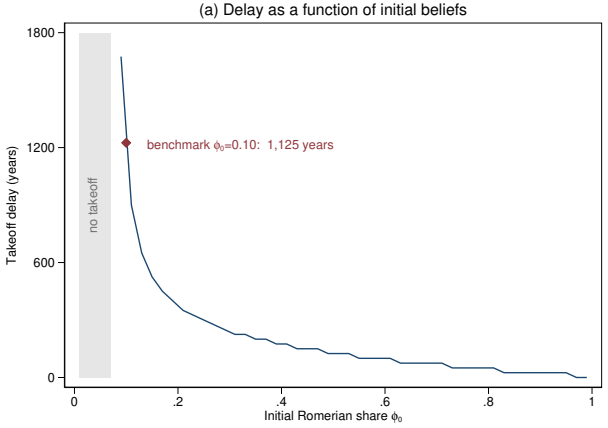
sensitivity




- $\phi_0 = 0.30$: fast takeoff (generation 20, ∼500 years)
- $\phi_0 = 0.10$: gap contracts to 0.09, composition crosses $p^*$ at gen. 20; takeoff at gen. 55 (∼1,375 yrs)
- $\phi_0 = 0.02$: trapped — credence converges to $< 1$, gap returns to zero

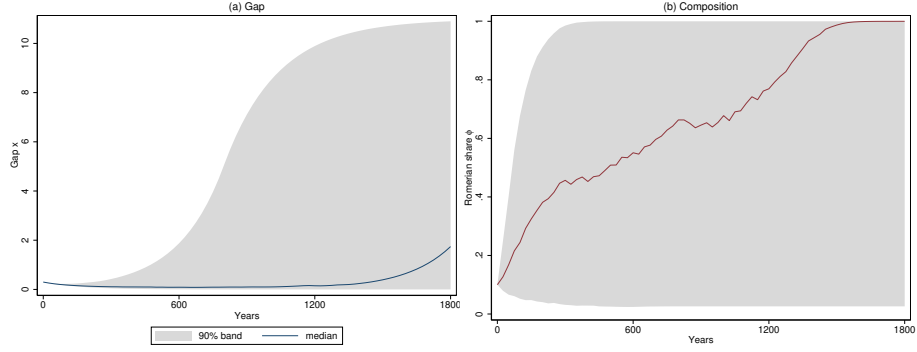
# Takeoff Delay and the Cost of Disagreement

welfare




- $\phi_0 \leq 0.07$: no takeoff within 2,000 years (numerical dormant trap)
- Delay falls steeply: 1,675 yrs at $\phi_0 = 0.09$; <300 yrs at $\phi_0 = 0.25$
- Welfare cost: $\omega \simeq 0.74\%$ of consumption per generation (20-gen window)

# Monte Carlo: The Role of Luck




- 500 replications, benchmark $\phi_0 = 0.10$; takeoff timing widely dispersed
- Fastest 5%: entry $<$500 years; bottom 5%: trapped forever

# Empirical Support

# Beliefs and Fertility: GSS Evidence

by education completed fertility

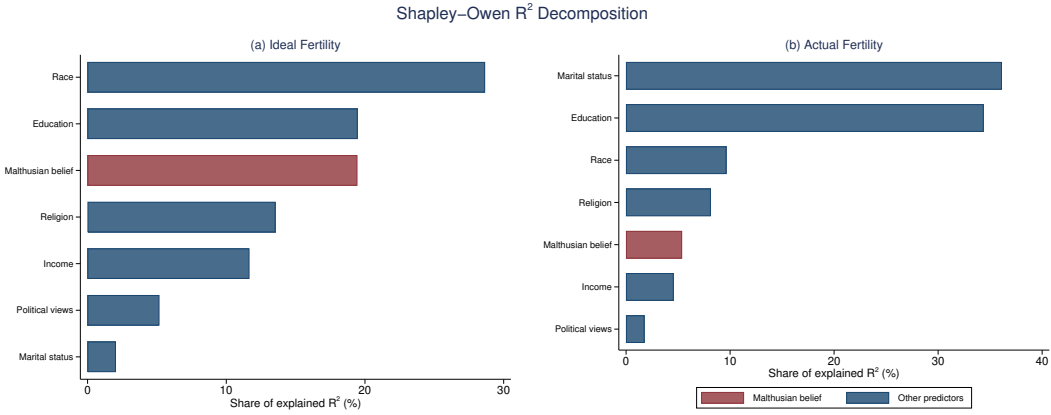
- GSS 2000 & 2010 ($N = 1{,}822$): agree/disagree "Earth cannot continue to support population growth" — 55% Malthusian

| | Ideal Fertility | | Actual Fertility | |
|---|---|---|---|---|
| | Baseline | Full | Baseline | Full |
| Malthusian | $-0.268^{***}$ | $-0.203^{***}$ | $-0.292^{***}$ | $-0.237^{***}$ |
| | (0.041) | (0.045) | (0.066) | (0.071) |
| Demographic controls + Year FE | Yes | Yes | Yes | Yes |
| Additional controls | No | Yes | No | Yes |
| Observations | 1,822 | 1,567 | 1,822 | 1,567 |
| $R^2$ | 0.060 | 0.099 | 0.262 | 0.333 |

- Baseline: age, age$^2$, sex, education, year FE
- Full adds: marital status, race, religion, political views, income

# Explanatory Power

compare



# Extensive Margin and Reverse Causality

details age groups

Margin decomposition:

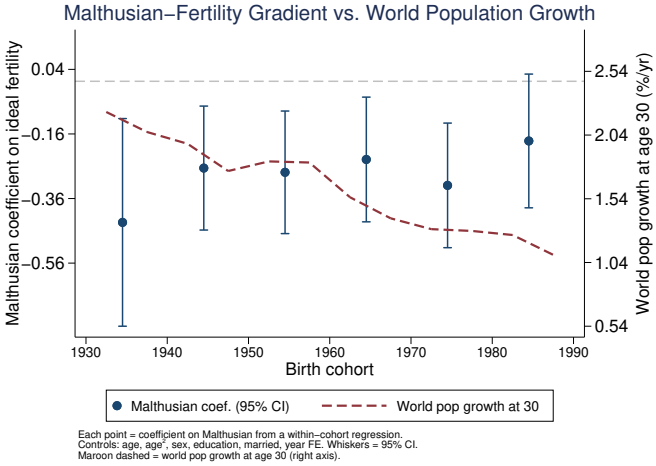
- Extensive: Malthusian odds ratio = 0.55 (AME: $-9.2$ pp, $p < 0.001$)
- Intensive (conditional on $> 0$): $-0.12$ ($p = 0.089$)
- Association concentrated on the decision to become a parent

Reverse causality checks:

- Ideal-fertility gradient significant in all age groups, including 18–34 (limited childbearing experience)
- Actual-fertility: zero among young, large among 35+
- Siblings control: coefficient changes $<5\%$

- Adding general environmental concern does not affect the results
- Pattern more consistent with beliefs $\rightarrow$ fertility

# Cohort Heterogeneity




- Negative correlation between the Malthusian coefficient and world population growth rate at the prime age

# Complementary Evidence: Malthus at Haileybury

- Robertson (2026): British Indian officials trained at Haileybury, where Malthus himself taught political economy
- Malthus's abrupt death $\Rightarrow$ replaced by Richard Jones (a critic) $\Rightarrow$ quasi-exogenous variation in doctrine across cohorts
- Officials taught by Malthus provided systematically less famine relief
- Three features: exogenous assignment, costly behavior (not cheap talk), decades-long durability
- Complements GSS: clean identification of the belief channel, though in bureaucratic relief rather than fertility

# Conclusion

- Disagreement about the sign of population scale effects generates:
  - → state-dependent fertility rankings
  - → boom-bust cycles in Malthusian worlds
  - → information traps in in Romerian worlds
- Pessimism delays takeoff: inherited beliefs contract the gap and starve future generations of evidence

# Ecological vs. Economic Malthusianism

back

- GSS question is in environment modules: captures ecological concern (resource depletion, sustainability)
- For the fertility margin, distinction is second-order: $A_t$ is broad TFP, and both idioms point to $\theta < 0$, a poorer future for descendants
- Including a measure of general environmental concern leaves the Malthusian coefficient essentially unchanged
- For historical claims the distinction is more substantive; the survey only validates the belief-to-fertility arrow

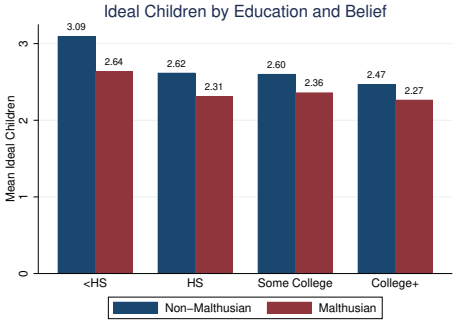
# Regression Evidence: Continuous Scale and Counts

back

| | Ideal Fertility | Actual Fertility |
|---|---|---|
| Malthusian (binary) | $-0.268^{***}$ | $-0.292^{***}$ |
| | (0.041) | (0.066) |
| Malthusian (1–5 scale) | $-0.153^{***}$ | $-0.146^{***}$ |
| | (0.021) | (0.033) |
| Poisson (binary) | $-0.107^{***}$ | $-0.179^{***}$ |
| | (0.016) | (0.037) |
| Excl. neutrals (binary) | $-0.357^{***}$ | $-0.342^{***}$ |
| | (0.054) | (0.087) |
| Controls + Year FE | Yes | Yes |
| Observations | 1,822 | 1,822 |

# Gradient Holds Within Education Groups

back




| | Ideal | | Actual | |
|---|---|---|---|---|
| | HS or Less | College+ | HS or Less | College+ |
| Malthusian | $-0.318^{***}$ | $-0.235^{***}$ | $-0.209^{*}$ | $-0.392^{***}$ |
| | (0.069) | (0.049) | (0.115) | (0.081) |

# Completed Fertility

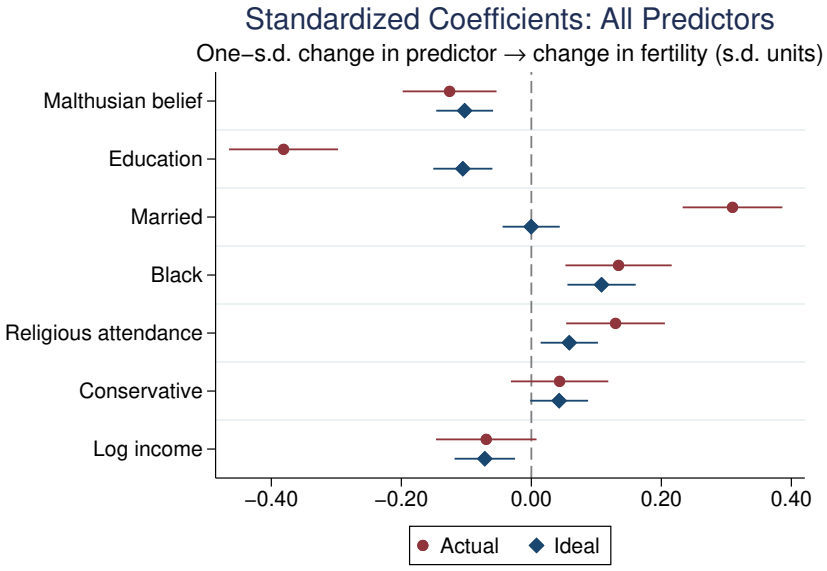
back

| | Full Sample | Completed | Not Yet Completed |
|---|---|---|---|
| Malthusian | $-0.292^{***}$ | $-0.257^{**}$ | $-0.288^{***}$ |
| | (0.066) | (0.109) | (0.079) |
| Observations | 1,822 | 848 | 974 |

- Completed = women 45+, men 50+; all controls included
- Estimate stable: not driven by incomplete childbearing among the young

# Normalized Coefficients

back



# Extensive vs. Intensive Margin

back

| | Extensive: Any Children | | Intensive: Children $\|> 0$ | |
|---|---|---|---|---|
| | Baseline | Full | Baseline | Full |
| Logit (OR) | $0.554^{***}$ | $0.533^{***}$ | | |
| | (0.067) | (0.080) | | |
| AME | $-0.092^{***}$ | $-0.080^{***}$ | | |
| | (0.019) | (0.019) | | |
| OLS | | | $-0.122^{*}$ | $-0.085$ |
| | | | (0.072) | (0.080) |
| $N$ | 1,822 | 1,567 | 1,299 | 1,120 |

Association concentrated on the decision to become a parent.

# Reverse Causality Checks

back

| | Full Sample | Age 18–34 | Age 35–49 | Age 50+ |
|---|---|---|---|---|
| *Panel A: Ideal Fertility* | | | | |
| Malthusian | $-0.268^{***}$ | $-0.243^{***}$ | $-0.322^{***}$ | $-0.253^{***}$ |
| | (0.041) | (0.075) | (0.079) | (0.061) |
| *Panel B: Actual Fertility* | | | | |
| Malthusian | $-0.292^{***}$ | $-0.052$ | $-0.469^{***}$ | $-0.263^{**}$ |
| | (0.066) | (0.094) | (0.124) | (0.114) |
| $N$ | 1,822 | 514 | 541 | 767 |

- Ideal gradient present even among young adults with little childbearing experience
- Actual: zero among 18–34, large among 35+ $\longrightarrow$ consistent with beliefs affecting fertility

# Why Each Preference Ingredient Is Load-Bearing

back

- Time cost + CRRA: TFP level cancels; income-supports-children channel off by construction — beliefs enter only via growth forecasts
- $\rho < 1$: required for positive dynastic value; $\rho > 1$ (with $\varepsilon > 1$) flips gradient sign — rejected by data
- $\varepsilon \in (0, 1)$: total weight on children increases with $n$ at diminishing rate; well-posed fertility problem
- Imperfect transmission: $\delta\lambda$ discounting in the fertility differential; association between parent's type and distant descendants decays
- Replacement normalization: $\beta^\dagger$ set so $n^0 = 1$; implied $\delta \simeq 0.253$

# Local Equilibrium Response: Details

back

- The local gap coefficient solves:

$$\big(a(p) - (1-\mu)\big)\big(1 - \delta a(p)\big) = K\bar{\theta}_p$$

- Explicit root: $a(p) = \frac{1+\delta(1-\mu)-\sqrt{[1-\delta(1-\mu)]^2-4\delta K\bar{\theta}_p}}{2\delta}$
- Individual policy slopes: $\left.\frac{\partial n_j}{\partial x}\right|_0 = b(p) + \frac{K(\theta_j - \bar{\theta}_p)}{1-\delta\lambda a(p)}$
- Amplification threshold: $a(p) = 1$ when $\bar{\theta}_p = \bar{\theta}^{dest} = \frac{\mu(1-\delta)}{K}$
- At baseline: $\bar{\theta}^{dest} \simeq 0.224$, requiring Romerian share $p^* \simeq 0.58$

# Dormant Trap: Formal Statement

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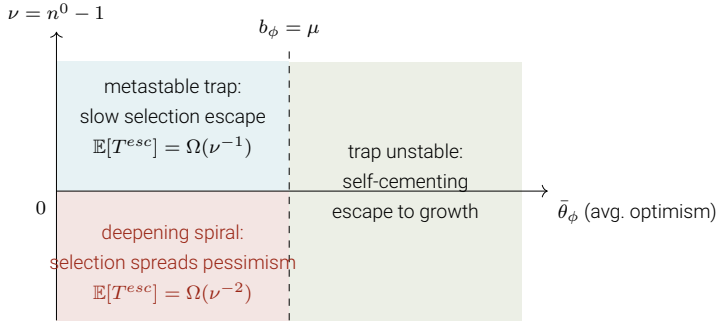
- Require uniform local return: $|F_x(0, \phi, \pi)| \leq q_0 < 1$ throughout a belief neighborhood
- On no-exit event $E$ (probability $\geq 1 - C|x_0|/d$):

$$|x_t| \leq q^t |x_0|, \qquad \sum_{t=0}^{\infty} \mathcal{I}_t \leq \frac{\Delta\theta^2 x_0^2}{2\sigma^2(1-q^2)}$$

- Limiting credence is close to initial: $\mathbb{E}[\mathbb{1}_E |\pi_\infty - p|] \leq C|x_0|$
- With $\lambda < 1$: adult composition converges to $\pi_\infty$
- Population converges to a positive finite level
- Long-run average log-productivity growth equals zero a.s. on $E$

# Trap Regimes Diagram

back




- $\nu > 0$: selection slowly favors Romerians; eventual escape
- $\nu < 0$: selection favors Malthusians; escape time quadratic in $|\nu|$

# Damped Cycles and Periodic Orbits

back

- Corollary (Damped demographic cycles): under uniformly negative $a(p) \in (-1, 0)$:

$$\operatorname{sgn}(x_{t+1}) = -\operatorname{sgn}(x_t), \qquad |x_t| \leq q^t |x_0|$$

  Population growth alternates; cumulative info finite; posterior interior

- Proposition (No periodic orbits under mean signals): replacing each signal by $s_t = \theta^* x_t$, no nonconstant periodic orbit with interior credence exists
  - → Every nonzero gap adds evidence for the truth
  - → Recurrent fixed-amplitude cycles require forgetting or outside shocks
- Damped-overshoot condition: $-\frac{(2-\mu)(1+\delta)}{K} < \bar{\theta}_p < -\frac{1-\mu}{K}$

# Computation Details

back

- Fertility implements first-order log-form response:

$$\ln n_{j,t} = \nu + K\left[\bar{\theta}_p X_t^1 + (\theta_j - \bar{\theta}_p) X_t^\lambda\right]$$

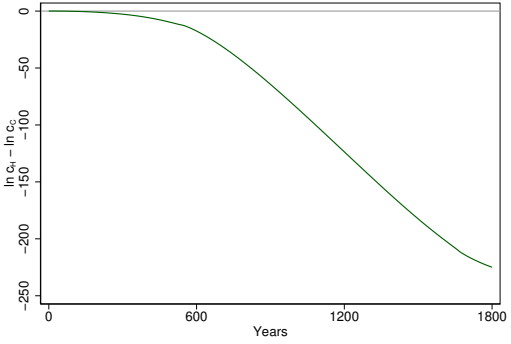
- $X_t^\lambda, X_t^1$ evaluated via constant-fertility forecast rule
- Aggregate fertility: within-period fixed point (local slope $\approx 0.07$, tolerance $10^{-12}$)
- TFR range: $[0.6, 6]$; beliefs capped at $(10^{-9}, 1 - 10^{-9})$
- Modern $\theta_R$ remark: semi-endogenous estimates (Jones 2022, Bloom et al. 2020) imply $\theta \approx 0.03 \ll \bar{\theta}^{dest}$ — even correct beliefs would not amplify
- Historical calibration uses strong-scale regime of Kremer (1993)

# Welfare Construction and Consumption Paths

back

Welfare measure:

- Dynastic consumption-equivalent over 20-generation window
- $\omega = (V_0^C / V_0^H)^{1/(1-\rho)} - 1$
- Benchmark: $\omega \simeq 0.74\%$ per generation
- Stable across $\lambda \times \sigma$ grid (0.71–0.75%)
- Conservative: $\delta \simeq 0.25$ per gen discounts distant future




Levels beyond the takeoff of both economies reflect the fertility cap and are illustrative only.

Log-consumption difference $H - C$: positive early (fewer children $\Rightarrow$ more time), negative and widening thereafter.

# Parameter Sensitivity

back

| Parameter variation | Delay (years) | Mechanism |
|---|---|---|
| $\lambda = 0.4$ | 950 | Faster recruitment to credence |
| $\lambda = 0.6$ (baseline) | 1,125 | — |
| $\lambda = 0.8$ | >1,775 | Family transmission insulates |
| $\sigma = 0.1$ | 175 | 4× more info per observation |
| $\sigma = 0.2$ (baseline) | 1,125 | — |
| $\sigma = 0.3$ | >1,775 | Too noisy to learn |
| $\theta_M = -0.075$ | 975 | Weaker contraction but less info |
| $\theta_M = -0.15$ (baseline) | 1,125 | — |
| $\theta_M = -0.30$ | 1,225 | Stronger contraction offsets more info |

- Transmission and noise interact strongly: high $\lambda$ + high $\sigma \Rightarrow$ censored
- $\theta_M$ moves delay modestly: contraction and info effects largely offset

# Robustness: What Would Break the Results

back

- No common crossing: rival models always distinguishable $\Rightarrow$ learning always active; trap vanishes
- $\rho > 1$: all rankings reverse, gradient flips — rejected by data
- $\mu \to 0$: trap reverts to slow learning; absorbing character requires $\mu > 0$
- Vanishing noise: trap dissolves if $\sigma^2(x)$ shrinks as fast as the squared gap
- Outside evidence: regional comparisons or research can identify $\theta^*$ without domestic demographic departure
- Persistent demographic shocks: can repeatedly displace economy into informative states, restoring learning
- Type-specific preferences: would preserve fertility differences at the reference $\Rightarrow$ exact dormancy breaks